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Abstract

This report will focus on a real-life cybersecurity incident in the automotive industry, specifically the 2014 hacking of the Uconnect infotainment system in a 2014 Jeep Cherokee. Vulnerabilities in connected software enabled the hackers to remotely take control of various components of the vehicle, such as the steering, brakes, transmission, and other in-car systems. This case brings to attention cyber-related threats and their significance for modern vehicles and passengers' lives (Miller & Valasek, 2015).

In terms of social issues, there was an apparent threat to human life. In addition, the incident affected people's trust in modern IoT vehicles. Moreover, privacy became an issue since data collected by the vehicle could be compromised. As for ethical issues, this case raised concern about a lack of consent for data collection, irresponsibility of hackers who reported on the incident, and misuse of the gained technical knowledge. Legally, this incident involved possible violations of the Computer Fraud and Abuse Act, product liability legislation, and data protection laws (Miller & Valasek, 2015).

Finally, the case highlighted the necessity of adhering to professional ethical standards. Specifically, the incident demonstrates the need to follow ethical guidelines set out by organizations such as IEEE and ACM.

Part 1: Introduction and Background of the Scandal

1.1 Introduction of the Company:

Figure 1: Fiat Chrysler Automobile (FCA)

Fait Chrysler Automobile (FCA) is an automobile **manufacturing company** that formed by the merger of Fiat **from Italy** and **Chrysler from USA** produces vehicles such as **the jeep Cherokee**. It developed the **Uconnect system**, a dashboard connectivity platform that provides **features** like **GPS navigation, radio, air conditioning control** and **wireless connectivity**. The company integrates **modern** technologies and **internet-based** systems into vehicles to enhance user **experience** and enable smart vehicle **functionality** (Elsaraf, 2021).

1.2 Introduction to Cyber Risks and Threats:

Cyber risk is defined in cybersecurity texts as the potential impact or damage that results when a cyber threat successfully exploits a vulnerability. It combines the likelihood of an attack with its potential consequences on systems, data, operations or safety (BAHMANOVA & LACE , 2024).

Cyber threats are defined as nay intentional action that can exploit vulnerabilities in digital systems to cause harm, steal data, disrupt operation or compromised integrity or availability of systems. It may include hacking, malware, phishing and other attack methods that target network, software etc (BAHMANOVA & LACE , 2024).

1.3 Different Types of Breaches:

i. **Remote Access Breach:** The hackers **Charlie Miller and Chris Valasek** were able to access Jeep Cherokee **remotely** over the internet from more than **10 miles** away without physical contact (Elsaraf, 2021).

ii. **Software Vulnerability Exploit:** The attackers discover **version 8.4AN** (WIFI and cellular network vulnerability) and **8.4A** (Touch Screen) **security flaws** that allowed unauthorized access in the Uconnect system (Elsaraf, 2021).

iii. **Network Breach (Internal System Access):** The attackers moved from the **infotainment** system into critical internal vehicle system (Elsaraf, 2021).

iv. **Unauthorized Control:** The attackers gained control over critical functions of a jeep Cherokee such as **engine, brakes, steering and GPS** (Elsaraf, 2021).

v. **Information Manipulation:** The Attackers **hacked** and manipulated systems like radio and GPS (Elsaraf, 2021).

vi. **IoT Security Risk;** The case highlights risk in internet of Things (IoT), where connected devices like **car** can be **controlled remotely** and become targets of cyberattacks (Elsaraf, 2021).

1.4 Background of the Scandal:

1.4.1 Overview of the Scenario:

In 2015, two cybersecurity researchers, Charlie Miller and Chris Valasek, conducted a historic "remote hack" of a 2014 Jeep Cherokee. They did this through the vehicle's Uconnect **infotainment** system. By exploiting security loopholes in the car's internet-connected software, they gained access to and control of critical systems, such as the **brakes**, **transmission**, and **steering**, in addition to non-critical systems like the radio and air conditioning, while the car was in motion. This was possible through the Uconnect system's cellular connection and specific **version vulnerability 8.4AN and 8.4A**, which acted as a gateway to the car's internal network, known as the **Controller Area Network** (CAN). The lack of proper authentication and open network ports enabled them to send messages to the electronic control units responsible for critical car systems. However, the 2015 "**Jeep hack**" demonstrated the possibility of accessing and controlling a car's systems without physical contact, which is a major security threat to cars (Mearian, 2015).

1.4.2 Past Similar Case:

Several Vehicle **hacking** incidents similar to the jeep Cherokee case have been reported over the years, highlighting serious cyber security vulnerabilities in the modern automobiles. Most of the hacked was successful due to the **unpatched software** or **bugs** in vehicle systems.

Case	Company	Year	Impact of the Attack	Cause
Toyota Prius and Ford Escape Hack.	Toyota, Ford	2013	Demonstrated that critical vehicle functions like brakes, and steering can be controlled, raising awareness of vehicle cyber security risks (Sachdev & Goyal, 2017).	By using cables to connect the devices to the vehicles through the on-board diagnostics (OBD)port (Kleinman, 2013). Click here
BMW ConnectedDrive Hack.	BMW	2015	Over 2.2 million Vehicles affected of remotely unlock, risk of unauthorized unlocking and theft (BBC, 2015).	Lack of encryption in communication (BBC, 2015). Click here
GM OnStar App Hack.	General Motors	2015	Vehicles could be remotely unlocked and started, risking theft and unauthorized access (Burden, 2015).	Weak authentication in mobile app (Burden, 2015). Click here
Nissan Leaf Hack	Nissan	2016	Remotely control all heating and cooling units (air, seat, steering wheel) (Garrad & Gilroy , 2021).	No authentication in API (Garrad & Gilroy , 2021). Click here
Tesla Model S Hack	Tesla	2016	Remote control of vehicle functions like brakes and doors (Nie & Du , 2016) .	Tesla Web Browser vulnerability (Nie & Du , 2016). Click here

Table 1: Past Similar Case

1.4.3 Intention of the Test Case:

The intention of the Charlie Miller and Chris Valasek were not to harm and rather their goal was to highlight serious cybersecurity vulnerabilities in connected vehicles so manufacturers would address them. By revealing how easily a vehicle could be compromised from the miles way, they aimed to push the automotive industry toward stronger security practices.

Video Click here

1.4.4 Impact if Done Unethically:

If such a breach were performed by **malicious attackers** rather than ethical researchers, the consequences could range from theft to serious accidents or **loss** of **life**. Automobiles are cyber physical systems where software compromise can translate directly to physical danger (Villasenor, 2015). Although **Transportation Department's National Highway Traffic Safety Administration** (NHTSA) ordered the carmaker to submit to "rigorous" federal oversight, buy back some defective vehicles from owners and pay a **$105 million civil penalty**, the largest ever issued by the NHTSA (Greenemeier, 2015) .

1.4.5 "WH" Question Rules:

Who?

The cybersecurity researchers **Charlie Miller** a former NSA hacker and **Chris Valasek** security expert conducted the **remote access breach** (Mearian, 2015)**.**

- **What?**

A **WIFI** and **cellular** network vulnerability **8.4A and 8.4AN** in the jeep Uconnect system was exploited, allowing **remote control** of vehicle functions (Mearian, 2015).

- **When?**

The hack was **conducted** in **July 2015** at Black HAT USA, **Las Vegas** (Drozhzhin, 2015).

- **Where?**

The demonstration occurred on a **highway** in the **St. Louis, Missouri** area while a wired magazine **journalist Andy Greenberg** was **driving** the vehicle (Jackson, 2015).

Part 2: Social Issues

The Phrase **"social issue"** is commonly employed **interchangeably** with the phrase **"social problem."** Social issue, a state of affairs that **negatively** affects the **personal** or **social** lives of individuals or the well-being of communities or larger groups within a society and about which there is usually public disagreement as to its **nature**, **causes**, or **solution** (Kulik, 2026).

The **jeep Cherokee hacking** incident highlighted but not technical **vulnerabilities** but also significant **social issues** related to cybersecurity, **safety** and **trust**. As vehicles becomes more connected through the **Internet of Thing** (IoT), they introduce new risks that affect **individual, society** and **industries**. These issues raise concerns about **privacy**, **safety, ethics** and the responsibility of companies to protect users from the **cyber threats**.

The **five social issues** may face by the both hacker and Fait Chrysler Company are:

2.1 Safety Risks to Human Life:

One of the most serious issues is the threats to passenger safety. Hackers **gaining** control of critical systems such as **brakes** and **steering** can cause **accidents**, **injuries** or even **loss of life**. This shows that **cyberattacks** on vehicles can directly **impact** physical safety of **human being** (Checkoway & McCoy, 2015) .

2.2 Privacy Concerns:

The connected vehicles collect and transmit data such as **location**, **driving**, **behaviour** and **personal information**. If hacked this data can be **accessed** or **misused** and violating **privacy**. These shows clear lack of **security issue** in automobile **industries** regarding **user information** and **privacy** (Checkoway & McCoy, 2015).

2.3 Loss of Trust on Technology:

The incident like **jeep Cherokee** hack case **reduces** public confidence in **smart** and connected **technologies**. Customers may become hesitant to **adopt advanced vehicle** features due to lack of **security flows** and the fear of **cyberattacks** (Checkoway & McCoy, 2015).

2.4 Risk to Public Confidence in Safety System:

Although the jeep hack was done by **ethical hackers Charlie Miller and Chris Valasek**, it raises questions about whether such demonstrations should be performed on public roads. The demonstration show risk of public if anything happens during hack on the public roads. There is a fine line between ethical research and **potential harm** (Miller & Valasek, 2015).

2.5 Public Fear and Anxiety:

The Jeep Cherokee hack created fear among the public about the safety of connected vehicles. People became concerned that hackers could remotely control important vehicle functions and cause accidents. This reduced public confidence in smart vehicle technologies and IoT systems (Greenberg, 2015).

Part 3: Ethical Issues

Ethics is a branch of **philosophy** that deals with what is considered to be **right** or **wrong**. Ethical issues are to understand **security**, **privacy issues** and **major negative** impacts of IT on cyberspace. The common **ethical problems** many individuals and organizations faced with, the issue of a company legally monitoring **employee's e-mail** is very controversial issues (Gunarto, 2026).

Ethical issues in cybersecurity involve concern about **privacy**, **unauthorized access** and **responsibility** for protecting systems. According to **Hary Gunarto** an **Indonesian computer engineer, scientist, writer** and **researcher,** accessing systems without permission is unethical even if done for research purposes (Gunarto, 2026). In the **jeep Cherokee case**, researchers remotely controlled a vehicle, raising concerns about user consent and **public safety**. **Privacy** is also a **major issue**, as connected vehicles collect sensitive data that could be **misused** if hacked. Organizations have an ethical responsibility to secure their systems and prevent such vulnerabilities.

The **five ethical issues** may face by the hackers and Fait Chrysler company are:

3.1 Unauthorized Public Safety Risk:

In the case of jeep Cherokee, the experiment was conducted on moving vehicle putting the Andy **Greenberg senior writer, journalist** and **Dr**. **Charlie Miller**, **Chris Valasek** self in life risk (Greenberg, 2015).

3.2 Lack of Consent:

In the following jeep case, the driver **Andy Greenberg** was not **fully** in control of the situation during the **hack** by Charlie Miller and Chris Valasek. Again, it is **unethical** to risk own and other life for the research (Greenberg, 2015).

3.3 Lack of responsibility Disclosure:

The Researcher **Charlie Miller** and **Chris Valasek disclosed** the vulnerability to the company before **public release**. It shows intention of **hacking** was to help the company to enhance **security flow** in the jeep Cherokee (Kleinman, 2013).

3.4 Unauthorized System Exploitation:

The same hacking knowledge of **Charlie Miller** and **Chris Valasek** can be used for both **protection** and **malicious** attacks to the jeep and cause the **harm** to the company (Greenberg, 2015).

3.5 Lack of Company Responsibility:

The following **test case** of **jeep Cherokee** shows that manufactures **failed** to secure their **systems** before releasing **car** them to the customers. The incident can **negatively** impact and reduce the trust of **customers** in the **automobile industries** (Miller & Valasek, 2015).

Comparison with Ethical Theories:

Ethical Issue	Utilitarianism Ethics	Deontology Ethics	Virtue Ethics	Right - Based Ethics
Unauthorized Public Safety Risk	Risking lives during the experiment is unethical if the harm caused is greater than the public benefit gained.	Professionals have a moral duty to protect human life and avoid actions that may endanger others.	A responsible professional should demonstrate care, responsibility, and good judgment during security research.	Every individual has the right to safety and protection from unnecessary harm or danger.
Lack of Consent	Conducting research without full consent may create more harm than benefit to the individuals involved	Ethical actions must respect individual rights and informed consent before conducting experiments.	Professionals should act honestly and respectfully toward all participants involved in research.	Every person has the right to make informed decisions about their own safety and participation.
Lack of Responsible Disclosure	Responsible disclosure benefits society by allowing vulnerabilities to be fixed	Professionals have a duty to report vulnerabilities responsibly to prevent public harm.	Reporting security flaws properly demonstrates honesty, responsibility,	Users have the right to secure systems and protection from avoidable cyber threats.

	before misuse occurs.		and professionalism.	
Unauthorized System Exploitation	Unauthorized exploitation is unethical if it creates fear, damage, or risks to society.	Accessing systems without permission violates moral duties and professional rules.	Responsible professionals should use their technical skills honestly and for beneficial purposes only.	Organizations and users have the right to security, privacy, and protection from unauthorized access.
Lack of Company Responsibility	Companies should ensure system security to protect the greatest number of users from harm.	Manufacturers have a duty to provide safe and secure products to customers.	Responsible companies should demonstrate accountability and commitment toward user safety.	Customers have the right to use secure products without fear of cyber threats or harm.

Table 2: Comparison with Ethical Theories

Part 4: Legal Issues:

Legal issues refer to any matter that may arise in the context of the law, including disputes, conflicts, and violations of laws or regulations (Anderson, 2023).

The jeep Cherokee hack raised serious legal concerns regarding cybersecurity, product safety liability. As Vehicles become connected systems, cyberattacks can lead to physical harm, making them to laws related to negligence, data protection and unauthorized access. The **hackers** and **Fait Chrysler Automobile** may face legal issues.

There are **five legal issues** may face by hackers and the manufacturers Fait Chrysler Automobile:

4.1 Computer Fraud and Abuse Act (CFAA – USA, 1986):

This law makes it illegal to access computer systems without **authorization**. In the jeep Cherokee case, **remotely accessing** the Uconnect system and taking control of vehicle functions would clearly **violate** this act (Checkoway & McCoy, 2015).

4.2 National Highway Traffic Safety Administration (NHTSA):

NHTSA is responsible for vehicle safety standards in the **United States**. When the vulnerability was discovered, Faid Chrysler had to recall around **1.4 million** vehicles to fix the security issue. The failure to address such safety risks lead to regulatory penalties **$ 105 Million** against the manufacturer (Greenemeier, 2015).

4.3 Product Liability Law (Consumer Protection Law):

Under product liability **law**, manufacturers and system developers including **Uconnect providers** have a legal duty to ensure their products are safe. If a defect in software leads to harm and the company can be held responsible for negligence, forced to pay settlements to affected **users**. (Checkoway & McCoy, 2015)

4.4 General Data Protection Regulation (GDPR – EU, 2018):

If such a system collects or transmits personal data like **location**, may breach of that data would **violate** privacy laws under **GDPR**. The Fait Chrysler automobile can face heavy fines if they fail to protect user data or do not implement proper **cybersecurity** measures (Checkoway & McCoy, 2015).

4.5 Tort Law:

Tort law allows victims to file civil lawsuits if negligence causes harm or financial loss. In this case, affected users could take legal action if insecure vehicle systems placed them in physical danger or caused damages (BBC, 2015).

Part 5: Professional Issues:

A **Profession** is a disciplined group of individuals who adhere to ethical standards and who hold themselves out as, and are accepted by the public as possessing special knowledge and skills in a widely recognised body of learning derived from research, education and training at a high level, and who are prepared to apply this knowledge and exercise these skills in the interest of others (Professions, 2003).

Professional issues are those issues that are related to the ethics, laws, and other aspects associated with any particular profession. These may include issues such as competency, confidentiality, conflict of interest, and other matters (Markbandiola, 2026).

The professional issues arise from experts not adhering to the **professional guidelines**. In cybersecurity, experts should ensure that there is responsible disclosure and no harm arises. The Jeep hack could lead to the occurrence of harm to people and even **violate** the **ACM** and **IEEE codes** if it were misused (ACM, 2018).

There are **five professional issues** may face to the hacker if intention was causing harm by misusing the vulnerability of the system.

5.1 IEEE Code of Conduct - Violation of Public Safety responsibility:

The primary duty of the professional is ensuring public safety. The exploitation of the vehicle control system could cause accidents or even loss of lives. According to ACM and IEEE Codes, professionals should not injure other people (IEEE, 2020).

5.2 ACM Code of Ethics - Failure of Responsible Disclosure:

Professionals are needed to follow responsible disclosure practice when conducting any cybersecurity work. Exploiting a flaw without first reporting it will go against the trust of cybersecurity (ACM, 2018).

5.3 ACM Professional Conduct - Misused of Technology Knowledge:

Technical knowledge can be used for protection or causing harm. Exploitation of technical vulnerabilities will breach the principle of professionalism as stipulated in the codes of conduct of BCS and ACM. (ACM, 2018)

5.4 IEEE Code of Ethics - Lack of Professional Integrity:

Integrity is among the key rules of being a professional. It includes doing what one is supposed to do and acting in good faith. Exploitation of a technical weakness will go against the principle of integrity and trustworthiness (IEEE, 2020).

5.5 ACM Code 2.3 - Organizational Policies, Laws and Regulations:

ACM Code 2.3 ("Know and Respect Existing Rules") states that professionals must follow laws, regulations, and organizational policies. Unauthorized exploitation of systems would violate both professional and legal responsibilities (ACM, 2018).

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6. Conclusion:

This report focuses on the Jeep Cherokee's Uconnect system got hacked. In particular, this incident illustrated the risks related to cyber threats for modern automobiles with built-in features of Internet of Things technologies. At the same time, this example illustrated the possible risks for people's lives, privacy, and reputation of connected vehicle manufacturers. Therefore, the following issues related to cybersecurity, including technical, social, ethical and professional were considered in this analysis.

It shows how important it is to ensure that a system is secure by proper design, patching of vulnerabilities on time, and implementing adequate access control mechanisms. The lesson further emphasizes the need for professionals and manufacturers to take appropriate action to ensure users' safety and security. In conclusion, the case is an important one in learning about IoT-based system security.

7. Personal Reflection:

This report may help them realise how deeply cybersecurity vulnerabilities can impact real lives. By identifying stakeholders such as vehicle owners, Fiat Chrysler Automobile (FCA), and researchers, they understood how one security gap affects many (Elsaraf, 2021).

Analysing failures in the Uconnect system made them appreciate the importance of ethical decision-making in technology. They now believe that professionals must follow responsible disclosure practices, conduct regular security assessments, and always prioritise public safety over convenience when building connected systems (Sachdev & Goyal, 2017).

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Appendix:

Dr. Charlie Miller Click Gmail Account

Chris Valasek Click Gmail Account

Figure 2: Charlie Miller and Chris Valasek

Target – 2014 Jeep Cherokee

The 2014 Jeep Cherokee was chosen because we felt it would provide us the best opportunity to successfully demonstrate that a remote compromise of a vehicle could result in sending messages that could invade a driver's privacy and perform physical actions on the attacker's behalf. As pointed out in our previous research [6], this vehicle seemed to present fewer potential obstacles for an attacker. This is not to say that other manufacturers' vehicles are not hackable, or even that they are more secure, but only to show that with some research we felt this was our best target. Even more importantly, the Jeep fell within our budgetary constraints when adding all the technological features desired by the authors of this paper (Miller & Valasek, 2015).

Click here.

Figure 3: Jeep Cherokee

Network Architecture:

The architecture of the 2014 Jeep Cherokee was very intriguing to us due to the fact that the head unit (Radio) is connected to both CAN buses that are implemented in the vehicle (Miller & Valasek, 2015).

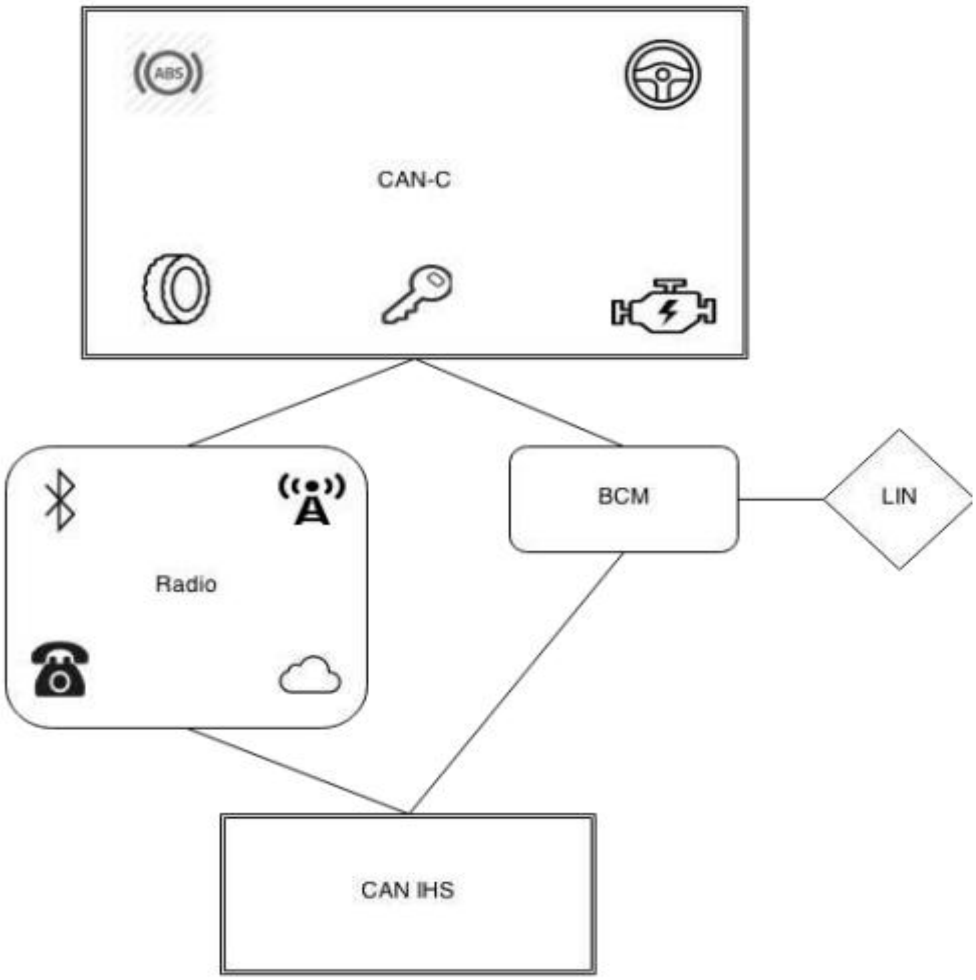

Figure 4: 2014 Jeep Cherokee Architecture Diagram

We speculated that if the radio could be compromised, then we would have access to ECUs on both the CAN-IHS and CAN-C networks, meaning that messages could be sent to all ECUs that control physical attributes of the vehicle. You'll see later in this paper that our remote compromise of the head unit does not directly lead to access to the CAN buses, and further exploitation stages were necessary. With that being said, there are no CAN bus architectural restrictions, such as the steering being on a physically separate bus. If we can send messages from the head unit, we should be able to send them to every ECU on the CAN bus (Miller & Valasek, 2015). Click here.

Overview of the Scenario: In Detailed

In 2015, cybersecurity experts Charlie Miller and Chris Valasek showed that it was possible to remotely hack into a Jeep Cherokee that was made in 2014 and was equipped with the Uconnect infotainment system. The experts were able to access the vehicle's internet-connected software and gain control of the vehicle's systems, including the brakes, transmission, and steering, even when the vehicle was in motion. The experts were able to do this because the vehicle's Uconnect system was connected to the vehicle's internal systems via the controller area network (CAN), which was connected to the vehicle's software via the vehicle's cellular connection. The vehicle's Uconnect system was connected to the vehicle's software via the vehicle's Wi-Fi and cellular connections. The vehicle's software was vulnerable because it had weak authentication and open ports that allowed the experts to gain access to the vehicle's internal systems and control the vehicle's systems.

Before this, Miller and Valasek had already remotely hacked vehicles such as a Toyota Prius and a Ford Escape in 2013, but they needed to have physical access to the car's diagnostic port to do so. This, however, was not the case with the 2015 Jeep hack, which proved that remote access without any physical touch was possible, thus increasing the threat level.

"WH" Question Rules:

- **Who**?

  Cybersecurity experts Charlie Miller, a former NSA hacker, and Chris Valasek, a security expert, carried out the remote hack.

- **What**?

The hackers took advantage of a vulnerability in the Jeep's Wi-Fi and cellular network connectivity through the Uconnect system.

- **When**?

The hack of the Jeep's car systems took place in July 2015, as demonstrated at Black Hat USA, and was made public shortly after.

- **Where**?

  The hack took place on a highway near the St. Louis, Missouri, region, where a journalist from Wired Magazine was driving the car.

- **Why**?

  The hackers wanted to highlight the importance of cybersecurity for car systems, as they were not being taken seriously.

- **How**?

  The hackers took advantage of the Jeep's Uconnect cellular connection and exploited open network ports and poor authentication.

What are 8.4AN and 8.4A?

- **8.4A:** Earlier version of Uconnect's 8.4-inch touchscreen system.
- **8.4AN:** "North America" variant of the 8.4A system, designed for US/Canadian vehicles.
- Both versions-controlled infotainment features (radio, GPS, climate control) and were connected to the internet via cellular and Wi-Fi.

Personal Reflection:

Furthermore, they were able to understand that today's cars are much more than just mechanical machines; rather, they have turned into sophisticated digital devices that communicate via networks and IoT technology. In this regard, cybersecurity is an important element to ensure safety as a software-related vulnerability can create threats not only to personal information but also to human lives. Through this insight, they became aware of the responsibility and role of cybersecurity experts and automobile companies in ensuring the protection of individuals from cybersecurity risks.

The report also gave them more insight regarding ethical hacking and proper approaches in security testing and research. In this sense, they discovered that ethical hacking has positive implications for society if done in an appropriate manner and vulnerabilities are addressed in advance. At the same time, they became familiar with the notion that improper use of technical information might cause fear, economic loss, accidents, and social consequences.

Overall, this research helped them develop a greater awareness of cybersecurity issues in connected technologies and think more thoroughly about ethical hacking and its potential consequences.